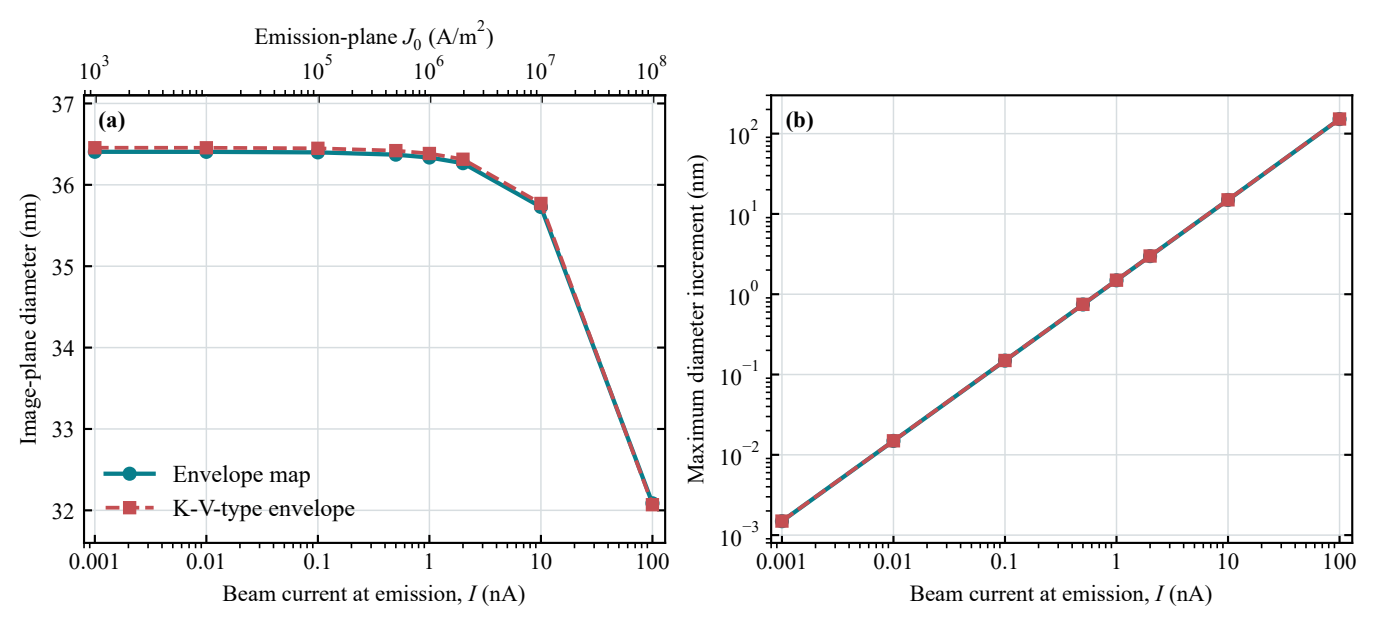

Emission-plane J_0 (A/m^2)
10^3
10^5
10^6
10^7
10^8
(a)
Image-plane diameter (nm)
37
36
35
34
33
32
Envelope map
K-V-type envelope
0.001
0.01
0.1
1
10
100
Beam current at emission, I (nA)
(b)
Maximum diameter increment (nm)
10^2
10^1
10^0
10^{-1}
10^{-2}
10^{-3}
0.001
0.01
0.1
1
10
100
Beam current at emission, I (nA)